WS01 Stream Geometry and Channel Survey Data

These datasets contain stream geometry, channel morphology, longitudinal elevation, and sensor-site width and depth measurements collected in Watershed 1 (WS01) at the H.J. Andrews Experimental Forest. Together, the four datasets describe spatial variation in channel structure and changes in wetted conditions along the study reach.

1. stream_geom_surveys.csv

Description: Repeated longitudinal surveys of stream geometry and hydrologic connectivity collected during 2024 and 2025. Surveys document changes in wetted channel dimensions, water depth, surface-water presence, flow status, and habitat type along the stream reach.

Unit of observation: Each row represents a surveyed point along the stream reach on a given date. Surveys collected across multiple days as part of the same sampling event are assigned a common survey date.

Variables:

- Date – Date the individual measurement was collected.
- Distance_m – Distance along the stream reach from the gage house (m).
- Wetted_width_m – Width of the wetted channel (m).
- Water_depth_m – Water depth (m).
- Wet_dry – Surface-water status; W = wet and D = dry.
- Flowing_notflowing – Flow status; F = flowing and NF = nonflowing.
- Riffle_run_pool – Habitat classification as riffle, run, or pool.
- survey – Standardized survey date used to group measurements collected over multiple days during the same longitudinal survey.

2. df_geomorphicfeatures.csv

Description: Single-reach survey characterizing geomorphic units, substrate type, and large wood along the study reach. Features originally recorded over distance ranges were expanded to 1-m longitudinal intervals.

Unit of observation: Each row represents a geomorphic observation at a specified distance along the stream reach. The dataset is in long format, so a given distance may have multiple rows corresponding to different feature categories.

Variables:

- Distance_m – Distance along the stream reach from the gage house (m).
- Category – Type of feature represented by the observation: Geomorphic_unit, Substrate_type, or Wood.
- Type – Classification associated with the corresponding category.

Classifications:

- Geomorphic units: SP = step-pool; PR = pool-riffle; BD = bedrock.
- Substrate: AL = alluvial; COL = colluvial; BD = bedrock.
- Wood: LJ = log jam; WD = woody debris.

3. df_slope.csv

Description: Longitudinal streambed elevation survey used to characterize the elevation profile and slope of the study reach.

Unit of observation: Each row represents a streambed elevation measurement at a specified longitudinal distance.

Variables:

- Distance – Distance along the stream reach (m).
- Streambed_elevation_m – Streambed elevation (m).

The longitudinal elevation profile can be used to calculate local changes in streambed elevation and reach-scale channel slope.

4. width_depth_surveys.csv

Description: Repeated measurements of wetted width and water depth at fixed sensor locations during 2024 and 2025. This dataset complements the full longitudinal surveys by tracking channel dimensions specifically at sensor locations.

Unit of observation: Each row represents a width and depth measurement at a sensor location on a given date.

Variables:

- Date – Date of measurement.
- site_position – Standardized sensor location identifier (S1–S17, including .a and .b site variants where applicable).
- Wetted_width_m – Width of the wetted channel at the sensor location (m).
- Water_depth_m – Water depth at the sensor location (m).

Relationship among datasets

The four datasets provide complementary information about the physical structure of the WS01 study reach. stream_geom_surveys.csv documents repeated reach-scale changes in channel dimensions and hydrologic connectivity; df_geomorphicfeatures.csv describes geomorphic, substrate, and wood characteristics; df_slope.csv provides the longitudinal streambed elevation

profile; and width_depth_surveys.csv provides repeated channel width and depth measurements at fixed sensor locations.